

Modelling and Forecasting Household Appliance Energy Use: A Comparative Time-Series Case Study

Time Series coding Case study and Report

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1. Introduction

Short-term forecasting of household appliance energy demand is directly relevant to smart-home energy management and demand-response scheduling. This report forecasts appliance energy consumption 24 hours ahead using the UCI Appliances Energy Prediction dataset, which combines appliance energy use with indoor sensor and outdoor weather readings recorded at 10-minute intervals over roughly four and a half months.

The central question is not simply which model achieves the lowest error, but whether additional complexity is justified by the improvement it delivers. Five model families are compared on an identical held-out test period: naive benchmarks, SARIMAX, a feature-based gradient-boosting model (XGBoost), and a pretrained time-series foundation model (Chronos), used zero-shot. The report covers the data, the forecasting problem definition, each model, the results, and a discussion answering the six assignment questions.

2. Data and Preprocessing

The dataset used is the UCI “Appliances Energy Prediction” dataset (Candanedo, Feldheim & Deramaix, 2017), downloaded directly from the UCI Machine Learning Repository. It contains 19,735 rows recorded at 10-minute intervals between 11 January 2016 and 27 May 2016, with 26 numeric columns: the target variable Appliances (Wh), lights energy use, nine indoor temperature/humidity sensor pairs (T1–T9, RH_1–RH_9), six outdoor weather variables (T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint), and two random “noise” columns (rv1, rv2) included by the original authors purely as a feature-selection sanity check, which were dropped before modelling since they carry no real information.

The raw data contained no missing values. To make SARIMAX fitting and grid search computationally manageable, the 10-minute series was resampled to hourly means, giving a working series of 3,290 hourly observations from 11 January 2016 17:00 to 27 May 2016 18:00. Resampling did not introduce any missing hours in this dataset, so no interpolation was required in practice, though the pipeline includes a time-based interpolation step as a safeguard for any small gaps.

2.1 Exploratory analysis

Figure 1 shows the full appliance energy series together with a one-week zoom. The series is highly spiky and right-skewed: most hours sit in a low baseline band (roughly 40–100 Wh), punctuated by sharp bursts corresponding to individual high-draw appliances switching on. A clear repeating daily rhythm is visible in the zoomed view, with activity concentrated in the morning and, more strongly, the evening.

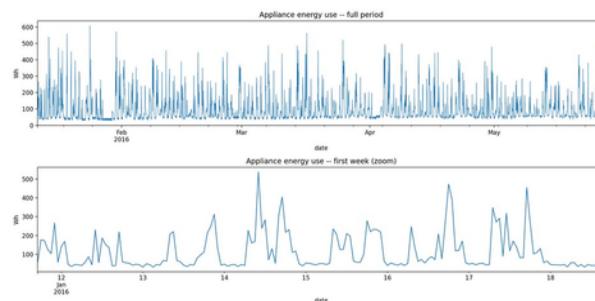


Figure 1. Appliance energy use — full period (top) and first week zoom (bottom).

An additive seasonal decomposition (period = 24 hours, Figure 2) confirms a stable daily seasonal component and a slowly drifting trend, with a residual component that retains a substantial amount of variance — consistent with the bursty, event-driven nature of appliance switching, which a smooth decomposition cannot fully capture.

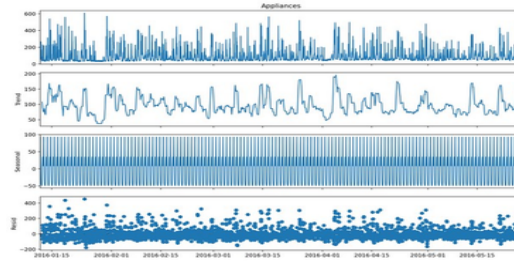


Figure 2. Additive seasonal decomposition of the hourly Appliances series (period = 24h).

2.2 Stationarity analysis

Stationarity was assessed using two complementary tests with opposite null hypotheses: the Augmented Dickey-Fuller (ADF) test (H_0 : unit root / non-stationary) and the KPSS test (H_0 : series is stationary). Agreement between the two gives more confidence than either alone. Both tests were run on the level series, after first-order differencing, and after first-order plus seasonal (24-hour) differencing. Results are summarised in Table 1.

Series	ADF stat.	ADF p-value	KPSS stat.	KPSS p-value	Conclusion
Level	-8.95	8.8e-15	0.038	0.10	Stationary
1st difference	-16.95	9.4e-30	0.078	0.10	Stationary
1st diff + seasonal diff (24)	-18.15	2.5e-30	0.081	0.10	Stationary

Both tests agree that the series is already stationary at the level, which at first appears surprising for a series with a visible daily rhythm. This is not a contradiction: ADF and KPSS test for a stochastic trend (a unit root) and long-run drift, not for the presence of seasonality itself. A series can have a strong, stable, repeating seasonal pattern and still be stationary in the ADF/KPSS sense, because the pattern does not change the series' long-run mean or variance. This is confirmed by the ACF plots (Figure 3): the level series shows a slowly decaying autocorrelation with a clear peak recurring every 24 lags, the signature of stable daily seasonality rather than a unit root.

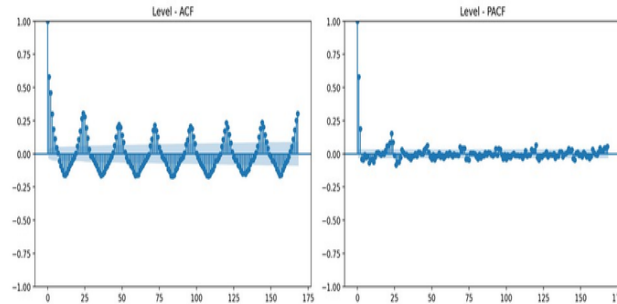


Figure 3. ACF and PACF of the level series, 168 lags.

Since ADF and KPSS already indicate stationarity at the level, no differencing is strictly required before fitting an ARMA-type model; however, differencing was still explored (Figures 4–5) to inform the SARIMAX seasonal order, since the ACF of the level series retains a strong seasonal signature at lag 24 that a purely non-seasonal ARMA specification would not capture well.

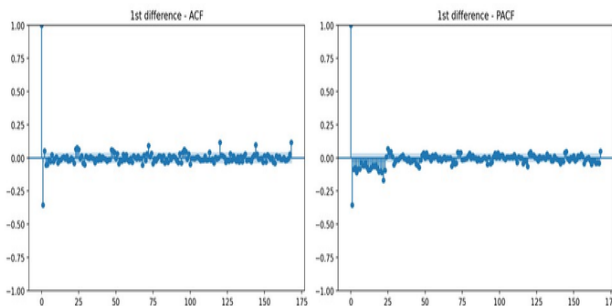


Figure 4. ACF and PACF after first-order (non-seasonal) differencing.

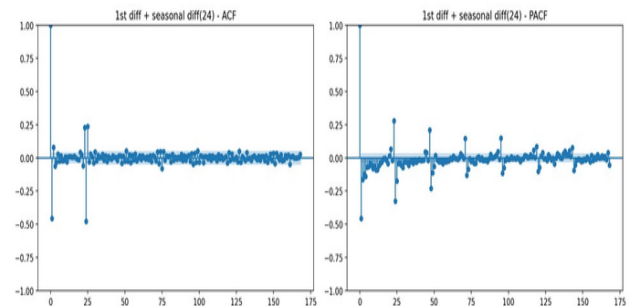


Figure 5. ACF and PACF after first-order plus seasonal (24h) differencing.

3. Forecasting Problem Definition

Element	Definition
Target variable	Appliances (Wh), hourly mean
Forecast horizon	24 hours (all models evaluated on the same forecast origin)
Train / test split	Chronological; final 14 days (336 hourly points) held out as test
Train period	11 Jan 2016 17:00 – 13 May 2016 18:00 (2,954 hourly observations)
Test period	13 May 2016 19:00 – 27 May 2016 18:00 (336 hourly observations)
Evaluation metrics	MAE, RMSE, MASE (scaled by in-sample seasonal-naïve error, seasonality = 24), Bias

All eight models (five benchmarks, SARIMAX, the feature-based model, and the foundation model) are evaluated on exactly the same 336-hour test window using the same four metrics, so that comparisons are on equal footing. MASE is used as the primary ranking metric because it is scale-free and directly interpretable relative to a seasonal-naïve baseline (MASE < 1 means the model beats seasonal-naïve forecasting, in-sample); RMSE and MAE are reported for completeness and because RMSE is more sensitive to the large occasional spikes that characterise this series; Bias indicates systematic over- or under-forecasting.

4. Benchmark Models

Five simple benchmark forecasts were implemented, each producing a full 24-hour-ahead (in this case, 336-hour, i.e. the whole test period in one shot) forecast from the training data only:

- Mean — the historical training mean, repeated for every future step.
- Naïve — the last observed training value, repeated (a random-walk forecast).
- Seasonal naïve (daily) — $\hat{y}(t) = y(t - 24)$, recursively extended past the first 24 steps.
- Seasonal naïve (weekly) — $\hat{y}(t) = y(t - 168)$, recursively extended.
- Drift — the naïve forecast extended by the average per-step change observed across the whole training series.

These represent the minimum bar that any more complex model must clear to justify its added complexity, and are discussed together with the full results in Section 8.

5. SARIMAX Model

A Seasonal ARIMA model with exogenous regressors (SARIMAX) was fitted to capture both short-run autocorrelation and the daily seasonal cycle identified in Section 2.2. The non-seasonal order (p, d, q) was selected by an exhaustive grid search over $p \in [0, 6]$, $d \in [0, 2]$, $q \in [0, 6]$ (147 combinations) using the Akaike Information Criterion (AIC), with the seasonal order fixed at $(P, D, Q, s) = (1, 1, 1, 24)$ based on the daily seasonality identified above.

For tractability, the search itself was run on the most recent 90 days (2,160 hours) rather than the full ~4-month training set, since fitting 147 full seasonal SARIMAX models on the complete series would take well over an hour; this is a deliberate, documented trade-off. The final model was refit on the entire training set (2,954 hours). Table 2 shows the top five candidate orders by AIC.

p	d	q	AIC
6	0	0	23084.81
0	0	6	23087.62
0	1	6	23091.17
5	0	0	23094.43
4	1	6	23096.00

The selected order was SARIMAX(6,0,0)(1,1,1,24) — a sixth-order autoregressive model with no differencing at the non-seasonal level, combined with a seasonally-differenced, seasonally-autoregressive and seasonally-moving-average daily component. Outdoor weather variables (T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint) were included as exogenous regressors, using their realised test-period values (see the important caveat on this in Section 9, Question 5).

Residual diagnostics (Figure 6, Table 3) show that while the AIC-selected order fits the training data well in likelihood terms, the residuals still exhibit statistically significant autocorrelation: a Ljung-Box test at lag 48 gives a p-value of 3.14×10^{-155} , decisively rejecting the null hypothesis of white-noise residuals. The residual distribution is also notably non-normal, with high positive skew (2.50) and heavy tails (excess kurtosis 12.90), reflecting the large, irregular energy spikes the linear

SARIMAX structure does not fully explain.

	Value
	888.85
Diagnostic	3.14e-155
Ljung-Box statistic (lag 48)	2.77
Ljung-Box p-value	71.74
Residual mean	2.50
Residual std. dev.	12.50
Residual skewness	
Residual excess kurtosis	

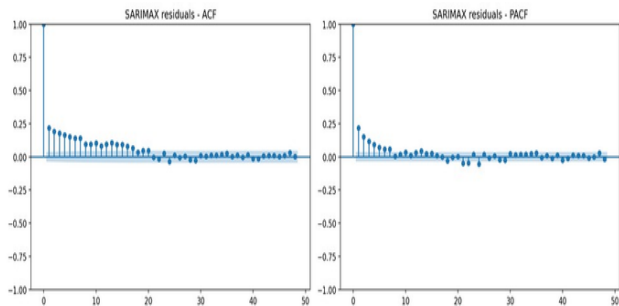


Figure 6. ACF/PACF of SARIMAX(6,0,0)(1,1,1,24) residuals.

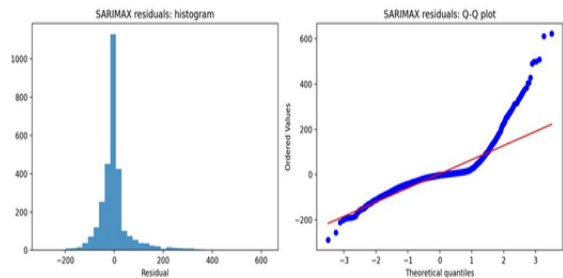


Figure 7. SARIMAX residual histogram and Q-Q plot.

6. Covariates and Feature-Based Model

6.1 Feature engineering

Three groups of predictors were engineered for the feature-based model, in addition to the raw indoor sensor and outdoor weather columns:

- Lag features: Appliances shifted by 1, 2, 3, 6, 12, 24, 48 and 168 hours.
 - Rolling features: rolling mean and standard deviation over 3, 6, 12, 24 and 168-hour windows, computed on the target after shifting by 1 hour so no window can include the current or a future observation.
 - Time features: hour-of-day and day-of-week encoded cyclically (sine/cosine), plus a weekend indicator.
- Data-leakage prevention was a specific design priority: every lag and rolling feature of the target is built with a one-step shift applied before any window is computed, so no feature can see the current or a future value of the series being predicted. This is verified directly by two of the project's automated unit tests (test_lag_features_do_not_leak_future_values, test_rolling_features_exclude_current_observation).

6.2 Model and results

An XGBoost gradient-boosting regressor (600 trees, learning rate 0.03, max depth 5, subsample and column-subsample 0.8) was trained on the full feature table — lag, rolling, time, indoor sensor, and outdoor weather features — and used to forecast the entire 336-hour test period directly from features computed at each test timestamp.

Feature importances (Figure 8, Table 4) show that recency-based features dominate: the single most important feature is the 1-hour lag (10.7% of total importance), followed by the 3-hour rolling mean (7.6%). Grouped by category, rolling-window features contribute the largest combined share of importance (25.0%), closely followed by lag features (23.4%), indoor sensor readings (22.4%) and time-of-day/day-of-week features (20.5%). Outdoor weather variables contribute comparatively little (7.6% combined), suggesting that for this dataset, the appliance series' own recent history and the time of day carry substantially more predictive signal than outdoor conditions.

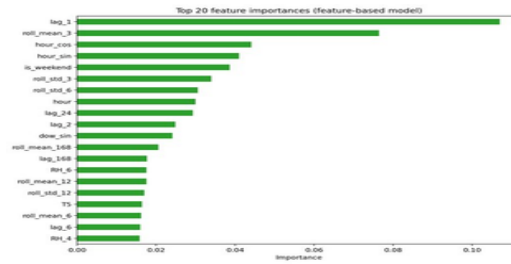


Figure 8. Top 20 XGBoost feature importances.

Feature group	Combined importance
Rolling (mean/std, target)	0.250
Lag (target)	0.234
Indoor sensor (T1-T9, RH_1-RH_9)	0.224
Time (hour, day-of-week, cyclic)	0.205
Outdoor weather	0.076
Other	0.011

7. Foundation Model (Chronos)

Amazon's Chronos (chronos-t5-small) was used as the time-series foundation model, applied in a genuine zero-shot setting: the pretrained model was given only the raw training values of the target series as context, with no fine-tuning, no covariates, and no feature engineering, and asked to forecast the 336-hour test period directly. Twenty sample trajectories were drawn and the median trajectory was used as the point forecast.

Two points are important for interpreting this result. First, Chronos is target-only by design in this configuration — unlike SARIMAX and the feature-based model, it was given no access to weather, sensor, or time-of-day information, so it competed on strictly less information. Second, the model itself warned during inference that “prediction length ≤ 64 is recommended” and that quality may degrade beyond that; since the test period requires a 336-step forecast, this is well outside Chronos's comfortable operating range, and some degradation in forecast quality over the second half of the horizon is expected as a consequence.

8. Results and Error Analysis

Table 5 summarises every model's performance on the identical 336-hour test period, ranked by MASE (the primary metric, scale-free and interpretable relative to seasonal-naïve).

Model	MAE	RMSE	MASE	Bias	% RMSE vs. best benchmark
XGBoost (feature-based)	32.70	54.77	0.612	+3.90	+26.9%
SARIMAX(6,0,0)(1,1,1,24)	36.10	64.23	0.676	-4.66	+14.3%
Chronos (foundation model)	40.28	79.42	0.754	-34.16	-6.0%
Seasonal naïve (weekly)	42.63	79.29	0.798	-10.82	-5.9%
Mean (strongest benchmark)	50.32	74.91	0.942	-3.11	0.0%
Seasonal naïve (daily)	86.96	129.23	1.628	+64.01	+72.5%
Naïve	250.64	258.82	4.692	+247.76	+245.5%
Drift	266.37	274.61	4.986	+264.50	+266.6%

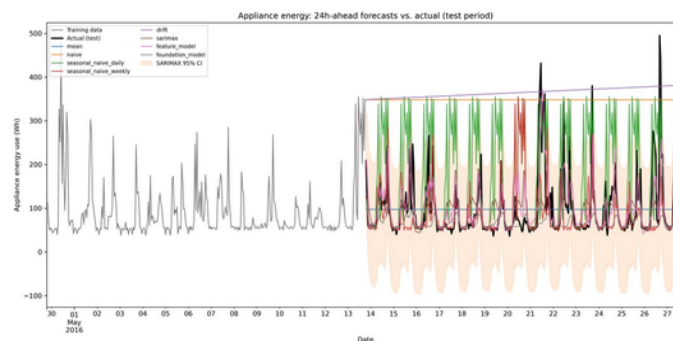


Figure 9. All model forecasts vs. actual values, test period, with SARIMAX 95% confidence interval.

Figure 9 shows every model's forecast against the actual test-period values. The two winning models, XGBoost and SARIMAX, both track the recurring daily double-peak pattern (morning and evening activity) reasonably well, though both under-predict the height of the largest, most irregular spikes (e.g. 21 and 27 May). The SARIMAX 95% confidence interval widens substantially over the horizon and, notably, extends below zero for large stretches — a physically impossible value for energy consumption — which is a direct symptom of fitting a linear Gaussian model to a strictly positive, right-skewed series with heavy-tailed residuals (consistent with the residual diagnostics in Section 5).

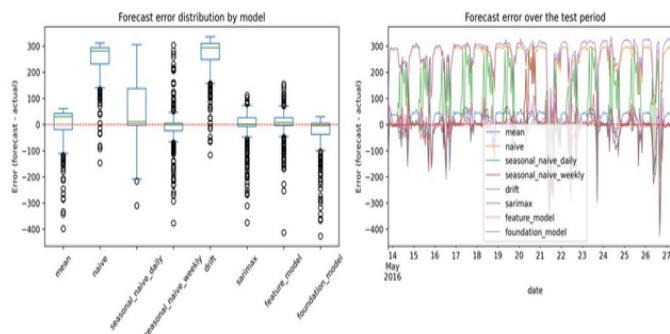


Figure 10. Forecast error distribution (left) and error over the test period (right), by model.

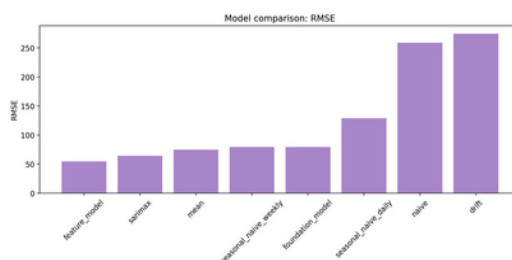


Figure 11. Model comparison by RMSE.

9. Discussion

This section directly addresses the six analysis questions set out in the assignment brief, drawing on the results presented above.

9.1 Which benchmark model is strongest, and what does this reveal about the structure of appliance energy use?

The plain historical mean achieved the lowest RMSE (74.91) and lowest MASE (0.942), narrowly ahead of seasonal-naive-weekly (RMSE 79.29, MASE 0.798). Seasonal-naive-daily was substantially worse (RMSE 129.23), and naive/drift were weakest by a wide margin. This ordering shows that appliance use does not follow a repeating daily template closely enough for “yesterday at this hour” to be reliable — individual days vary too much in when and how intensely appliances are used. That naive and drift (short-run momentum) perform worst suggests one high-usage hour does not reliably predict the next, while weekly-cycle structure (e.g. weekday/weekend routines) is present but noisy.

9.2 Does SARIMAX improve on the strongest seasonal benchmark? Are daily seasonality, autocorrelation, and exogenous variables adequately captured?

Yes: SARIMAX achieves RMSE 64.23 and MASE 0.676, a 14.3% RMSE improvement over the mean benchmark and a substantial improvement over every other benchmark. The AIC-selected order — a sixth-order autoregressive term with a seasonally-differenced daily component — is consistent with the ACF evidence (Section 2.2) of both short-run autocorrelation and a stable 24-hour cycle. However, “adequately captured” needs qualification: the Ljung-Box test rejects white-noise residuals overwhelmingly ($p \approx 3 \times 10^{-155}$), and the residuals are strongly non-normal (skew 2.50, excess kurtosis 12.50). In practice, SARIMAX captures the broad daily rhythm but systematically misses the sharp, irregular appliance-switching spikes driving the heavy right tail — exactly what a linear Gaussian model is least suited to represent. The exogenous weather variables' marginal contribution was not isolated via a separate ablation, noted as a limitation in Section 10.

9.3 Does the XGBoost model improve when lag, rolling-window, time-of-day, and sensor/weather

variables are added? Which feature groups appear most useful?

The feature-based model is the strongest overall (RMSE 54.77, MASE 0.612, +26.9% RMSE over the mean benchmark), demonstrating the added features carry real predictive signal beyond SARIMAX's purely autoregressive structure. The importance breakdown (Table 4) shows rolling-window features (25.0% combined) and lag features (23.4%) as the largest contributors — the single most important feature is the 1-hour lag. Indoor sensor readings (22.4%) and time-of-day/day-of-week features (20.5%) contribute similarly, while outdoor weather contributes comparatively little (7.6%). This suggests the model leans primarily on the target's own recent history and cyclical time structure, which makes physical sense: an appliance's recent draw and the time of day are more directly linked to whether it is running than outdoor temperature is.

9.4 Does the foundation model outperform the simpler benchmark, SARIMAX, and feature-based models? Is any improvement large enough to justify the extra complexity?

No. Chronos, run zero-shot with no fine-tuning and no covariates, achieved RMSE 79.42 and MASE 0.754 — worse than SARIMAX and XGBoost, and worse than the mean benchmark on RMSE. It also showed the largest bias of any model (−34.16), a strong under-forecasting tendency consistent with the model's own warning that the 336-step horizon requested is well beyond its recommended maximum of 64 steps. Two factors likely explain this: Chronos was given strictly less information than SARIMAX or XGBoost (no weather, sensors, or explicit time-of-day signal), so the comparison is not entirely like-for-like; and forecasting $\approx 5.5\times$ beyond the recommended length likely degrades quality substantially. Given the weaker accuracy and added inference cost, the foundation model's complexity is not justified here in its current zero-shot, target-only configuration. A fairer test would restrict it to its recommended horizon or use a covariate-aware variant (Section 10).

9.5 Which variables would genuinely be known at the forecast origin? Are the SARIMAX/feature-model forecasts true forecasts or conditional forecasts?

Time-of-day and day-of-week features are genuinely known in advance and usable without qualification. The target's own lagged and rolling-window values are also legitimately available, since they are constructed only from past observations (Section 6.1). However, the outdoor weather variables used as SARIMAX exogenous regressors and the indoor sensor readings used as XGBoost features were the realised test-period values, not values available at the moment a real forecast is

issued — a genuine deployment would only have a weather forecast (with its own uncertainty), and no future indoor readings at all. This means SARIMAX and the feature-based results reported here are conditional forecasts, and their accuracy should be read as an upper bound on what is achievable in practice. Since outdoor weather contributed only 7.6% of XGBoost's feature importance (Table 4), the practical impact is likely modest for the feature-based model, though it cannot be ruled out as material for SARIMAX, which has no equivalent importance breakdown.

9.6 Which model would you recommend for practical smart-home energy forecasting, and why?

Considering accuracy, interpretability, uncertainty, cost, and ease of deployment together, the feature-based XGBoost model is recommended. It achieved the best accuracy on every metric (Table 5), its importances are directly interpretable and physically sensible (Section 6.2), it trains and predicts in seconds rather than the tens of minutes needed for the SARIMAX grid search or Chronos's download/inference overhead, and it degrades gracefully to a dependency-free alternative if XGBoost is unavailable. Its main weakness is that it does not natively provide calibrated prediction intervals; if uncertainty quantification is a hard requirement, SARIMAX's confidence intervals — despite extending into physically impossible negative territory here — or a quantile-regression variant of the feature-based model would be worth the extra complexity. Chronos is not recommended in its current zero-shot, beyond-recommended-horizon configuration, though it remains worth revisiting if restricted to shorter horizons or extended to use covariates.

10. Limitations and Future Work

- The SARIMAX order was grid-searched on the most recent 90 days, not the full training set, for tractability; a full- data or stepwise search (Hyndman & Khandakar, 2008) might select a different order.
- SARIMAX and the feature-based model use realised test-period weather/sensor values (a conditional, not true operational, forecast — Section 9.5). A deployed pipeline would substitute genuine weather forecasts.

- Chronos was evaluated zero-shot, target-only, and well beyond its recommended 64-step length for a single 336-step forecast. A rolling-origin evaluation with shorter horizons, or a covariate-aware variant (Chronos-2), would be fairer.
- A single train/test split (final 14 days) was used rather than multiple rolling-origin folds; cross-validated evaluation would give a more robust performance estimate.
- SARIMAX residuals indicate the linear Gaussian assumption fits this heavy-tailed, skewed error distribution poorly; a model built for spike-heavy data (e.g. a quantile-based ensemble) may represent it better.
- An ablation of SARIMAX with/without exogenous weather variables was not run; this would quantify their value for SARIMAX, mirroring the importance analysis already available for XGBoost.

11. Conclusion

This case study compared five families of forecasting model for 24-hour-ahead household appliance energy demand. The strongest simple benchmark (the historical mean) was beaten meaningfully by both SARIMAX (+14.3% RMSE) and, more substantially, by a feature-based XGBoost model (+26.9% RMSE), confirming that structured modelling of this series' autocorrelation and covariates is worthwhile. A genuinely zero-shot foundation model (Chronos), by contrast, underperformed the simplest benchmark on RMSE, highlighting that off-the-shelf foundation models are not automatically superior to purpose-built models when used without covariates and beyond their recommended horizon. The feature-based model is recommended for practical deployment on accuracy, interpretability, and cost grounds, with the caveat that its evaluation reflects a conditional rather than fully operational forecast.

References

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